The High-Bar and Low-Bar Back-Squats: A Biomechanical Analysis

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Abstract

Glassbrook, DJ, Brown, SR, Helms, ER, Duncan, S, and Storey, AG. The high-bar and low-bar back-squats: a biomechanical analysis. *J Strength Cond Res* XX(X): 000–000, 2017—No previous study has compared the joint angle and ground reaction force (vertical force [*Fv*]) differences between the high-bar back-squat (HBBS) and low-bar back-squat (LBBS) above 90% 1 repetition maximum (1RM). Six male powerlifters (POW) (height: 179.2 ± 7.8 cm; bodyweight: 87.1 ± 8.0 kg; age: 21–33 years) of international level, 6 male Olympic weightlifters (OLY) (height: 176.7 ± 7.7 cm; bodyweight: 83.1 ± 13 kg; age: 22–30 years) of national level, and 6 recreationally trained male athletes (height: 181.9 ± 8.7 cm; bodyweight: 87.9 ± 15.3 kg; age: 23–33 years) performed the LBBS, HBBS, and both LBBS and HBBS (respectively) up to and including 100% 1RM. Small to moderate ($d = 0.2–0.5$) effect size differences were observed between the POW and OLY in joint angles and *Fv*, although none were statistically significant. However, significant joint angle results were observed between the experienced POW/OLY and the recreationally trained group. Our findings suggest that practitioners seeking to place emphasis on the stronger hip musculature should consider the LBBS. Also, when the goal is to lift the greatest load possible, the LBBS may be preferable. Conversely, the HBBS is more suited to replicate movements that exhibit a more upright torso position, such as the snatch and clean, or to place more emphasis on the associated musculature of the knee joint.

Key Words joint angles, ground reaction forces, powerlifting, Olympic weightlifting

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Introduction

The squat is one of the most common exercises in strength and conditioning. The movement is widely accepted as valid and reliable for the assessment and improvement of lower extremity/trunk strength, function, and resilience to injury (4,9,10) and an effective exercise in injury rehabilitation (19). These benefits are possible through the contributions of the quadriceps, hamstrings, gluteal, triceps surae, and lumbar erector muscle groups to the completion of the movement (9,25). In fact, it is predicted that more than 200 muscles are active throughout the completion of a single repetition (31,36). The squat itself is in essence a simple movement, despite the great number of active muscles throughout. In strength and conditioning, load can be applied to the squat movement via several methods, for example dumbbells, kettlebells, and a range of other weighted implements. However, perhaps most commonly, load is applied via a barbell and in one of 2 ways: (a) as a front-squat, where a barbell is placed anteriorly on the shoulder and (b) as a back-squat, where the barbell is placed posteriorly to the shoulder and across the trapezius musculature (16). The focus of this article will be the back-squat.

There are 2 different variations of the back-squat, differentiated by the placement of the barbell on the trapezius musculature. The traditional "high-bar" back-squat (HBBS) is performed with the barbell placed across the top of the trapezius just below the process of the C7 vertebra and is commonly used by Olympic weightlifters (OLY) to simulate the catch position of the Olympic weightlifting competition lifts, the snatch and clean and jerk (41). Conversely, the "low-bar" back-squat (LBBS) places the barbell on the lower trapezius, just over the posterior deltoid and along the spine of the scapula (41). The LBBS is commonly used in competitive powerlifting (where the back-squat is 1 of the 3 competition lifts), as it may enable higher loads to be lifted (32). This could be due to the maximization of posterior displacement of the hips and increased force through the hip joints in comparison with the knee joints (37). The differences in bar position between the HBBS and LBBS result in an altered center of mass. Therefore, movement strategies result to maintain the bodies' center of mass within its base

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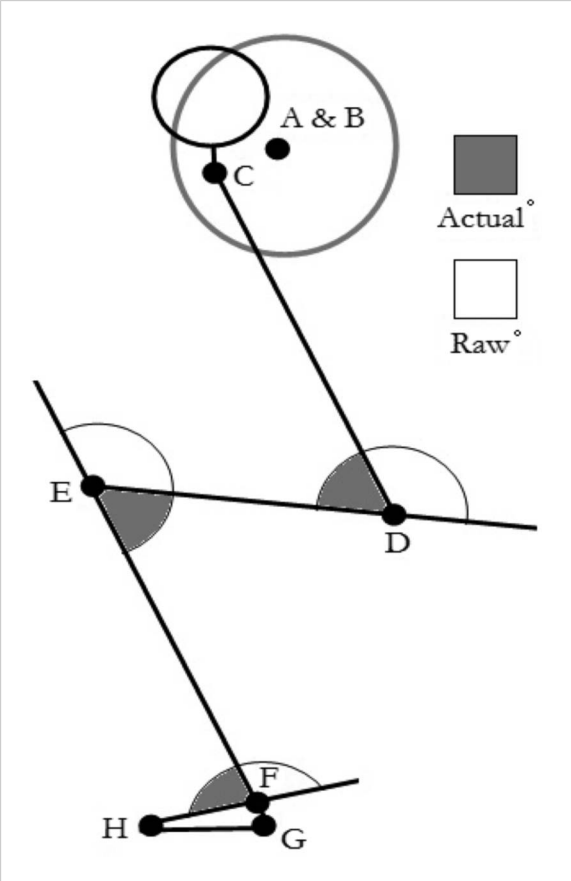

Figure 1. Actual and raw joint angles of the hip, knee, and ankle: taken from (A) the left end of the barbell, (B) the right end of the barbell, (C) acromion process, (D) greater trochanter, (E) lateral epicondyle of the femur, (F) lateral malleolus, (G) the top of the heal lift of the lifting shoe, and (H) the base of the fifth metatarsal.

of support (BOS). These movement strategies may manifest as: changes in (a) joint angles of the lower extremity kinetic chain and (b) ground reaction forces (vertical force [*Fv*]).

When comparing the HBBS with LBBS, several differences present themselves. In powerlifting, there are competition regulations that each lifter must comply with in order for each lift to count toward their competition total (21). One such regulation is for sufficient "depth" to be reached in the squat. That is, there must be sufficient flexion of the knees and lowering of the hips toward the ground, so that "the top surface of the legs at the hip joint is lower than the top of the knees" (21). In comparison, the HBBS is not directly included as a competition lift in Olympic weightlifting. Therefore, in training, OLY typically squat to a depth that replicates the final catch position of the snatch and clean and jerk. This often manifests as a deeper squat position than powerlifting regulation depth, characterized by greater flexion at the hip, knee, and ankle joints. Previous research has shown that the angle at peak knee flexion is generally smaller in the HBBS (e.g., 70–90°), in comparison with the LBBS (e.g., 100–120°) (5,11,13,17,18,20,24,27,37,38). Interestingly, some studies have reported the reverse (17,24,37). These conflicting results (although not explicitly stated by the authors) are likely to be the raw joint angles and not the actual angle (Figure 1).

Moreover, previous research specifically comparing the HBBS with the LBBS shows that the LBBS is defined by a smaller absolute trunk angle, and therefore greater forward lean to maintain the barbell over the center of mass (2,14,41). The unique position of the LBBS results in (a) a decreased trunk lever arm when placing the bar lower on the back, (b) a greater emphasis on the stronger musculature of the hip rather than the musculature of the knee joint, and (c) an increase in stability and a potential decrease in stress placed on the lumbar region and ankle, when compared with the HBBS (34,37). These factors may contribute to understanding why the LBBS typically allows for greater loads to be lifted. However, these kinematic findings are not definitive, and there are mixed results in the literature for the size of HBBS and LBBS trunk angles at peak hip flexion (5,11,13,17,20,24,27,29,37). Similarly, no conclusive differences between the HBBS and LBBS ankle joint angles can be drawn, in reference to previous literature (13,17,24,34,37).

As the position of the barbell on the trapezius influences the joint angles of the back-squat, there is also a resultant influence on the *Fv* produced. The position of the upper body (i.e., hip joint angle) has a large impact on the location and magnitude of the resultant *Fv* because of its larger mass. Because of the LBBS tending to allow for greater loads to be lifted, it would be expected that the *Fv* produced would be greater than that with the HBBS. However, the 2 studies that have specifically compared the *Fv* profiles of the HBBS and LBBS provide contradictory results to this expectation (15,37). The results of these 2 studies may indicate that, although the LBBS typically allows for greater load to be lifted through apparent mechanical advantages such as

Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
Familiarization session 1			Familiarization session 2		'Self-directed' familiarization session 3		Testing session

Figure 2. Representation of the order of familiarization and testing dates for the comparison group.

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Table 1. Mean loads lifted across all %1RM ranges.*†

% Range	Variable	OLY	POW	HBCOM	LBCOM	OLY vs. POW Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
	BW (kg)	83.2 ± 13.0	87.1 ± 8.0	87.9 ± 15.3	87.9 ± 15.3		
74–83	Load (kg)	136.6 ± 23.5	140.9 ± 20.1	99.9 ± 13.2	103.0 ± 16.2	12.5 ± 23.8	4.0 ± 7.6
	BW	1.6 ± 0.2	1.6 ± 0.3	1.2 ± 0.2	1.2 ± 0.2	0.1 ± 0.3	0.1 ± 0.1
84–93	Load (kg)	152.5 ± 23.1	159.2 ± 21.8	116.4 ± 12.9	121.7 ± 18.8	9.4 ± 26.6	6.0 ± 9.5
	BW	1.8 ± 0.2	1.9 ± 0.4	1.3 ± 0.2	1.4 ± 0.2	0.0 ± 0.3	0.1 ± 0.1
94–99	Load (kg)	164.0 ± 24.7	174.6 ± 20.1	128.7 ± 12.4	136.5 ± 21.6	7.2 ± 24.2	7.9 ± 8.0
	BW	2.0 ± 0.2	2.0 ± 0.4	1.5 ± 0.2	1.6 ± 0.2	0.0 ± 0.3	0.1 ± 0.1
100	Load (kg)	169.5 ± 26.5	181.2 ± 21.8	135.2 ± 11.1	143.4 ± 20.7	11.8 ± 25.4	8.2 ± 11.1
	BW	1.9 ± 0.3	2.1 ± 0.4	1.6 ± 0.2	1.6 ± 0.2	0.1 ± 0.3	0.1 ± 0.1

*1RM = 1 repetition maximum; BW = times body weight; CI = confidence interval; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters.
†All data are presented as mean ± *SD*.

a decreased trunk lever arm, these mechanical advantages are not effectively displayed by *Fv*. Furthermore, the results of these studies specifically may have arisen because of the level of expertise of the participant with performing the LBBS, as the authors chose to target the HBBS in recruitment as the focus for expertise. Therefore, further research is warranted to understand the *Fv* differences between the HBBS and LBBS, in particular with loads greater than 90% 1 repetition maximum (1RM).

The existing literature provides some insight into the kinematic and kinetic differences between the HBBS and LBBS. However, there is no consensus as to the differences between the 2 back-squat barbell positional variations. At present, no previous study has compared the joint angles and *Fv* of the HBBS and LBBS above 90% 1RM, and some results may have been confounded by inadequate familiarization. Thus, the purpose of this study was to compare and contrast the differences in joint angles and *Fv* of the HBBS and LBBS, up to and including maximal effort, in an effort to create a full profile of the 2 bar back-squat variations in groups both well versed and newly introduced to these movements. The results of this investigation will add to the current body of knowledge of Olympic weightlifting and powerlifting practice alike and provide an understanding of why the LBBS may allow for a greater load to be lifted.

Methods

Experimental Approach to the Problem

To determine why the LBBS may allow for greater loads to be lifted than the HBBS, both squat styles were performed by experienced and inexperienced lifters. The HBBS was performed by experienced OLY, and the LBBS by experienced powerlifters (POW), up to and including 100% of 1RM. Recreationally trained athletes served as a comparison group and performed both the HBBS and LBBS. It is assumed that the experienced OLY and POW have a better technique than the recreationally trained athlete; however, it is important to acknowledge this may not be strictly true in practice. A profile of each squat was created through analysis of kinematic joint angles and kinetic *Fv* differences.

Subjects

Six male POW (height: 179.2 ± 7.8 cm; bodyweight: 87.1 ± 8.0 kg; age: 21–33 years) of international (i.e., Oceania championships) level volunteered to participate in the LBBS group. In addition, 6 male OLY (height: 176.7 ± 7.7 cm; bodyweight: 83.1 ± 13 kg; age: 22–30 years) who had previously qualified for national championship–level competition volunteered to participate in the HBBS group. All POW routinely performed the LBBS in training and competition, and all OLY routinely performed the HBBS in training. Finally, 6 recreationally trained male athletes (height: 181.9 ± 8.7 cm; bodyweight: 87.9 ± 15.3 kg; age: 23–33 years) volunteered as a comparison group, and each

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Table 2. Mean loads lifted effect sizes and percentage differences.*

		OLY vs. POW		HBCOM vs. LBCOM	
% Range	Variable	Effect size	% Difference	Effect size	% Difference
74–83	Load (kg)	0.3†	3.2	0.3†	3.0
	BW	0.1	0.2	0.3†	2.5
84–93	Load (kg)	0.2†	4.4	0.3†	4.4
	BW	0.0	0.8	0.4†	3.9
94–99	Load (kg)	0.2†	6.5	0.5‡	5.7
	BW	0.0	2.5	0.6‡	5.1
100	Load (kg)	0.2†	6.9	0.4†	5.7
	BW	0.1	3.0	0.5‡	5.2

*BW = times body weight; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.

participant was required to perform both the LBBS and HBBS in a randomized order, after 2 familiarization sessions with both types of squat. All participants were free of injury and had ≥1 year's strength training experience (POW: 5.05 ± 4.56 years; OLY: 3.75 ± 2.72 years; recreational: 8.67 ± 3.5 years) consisting of ≥3 training sessions per week for the POW and OLY. The comparison group volunteers were required to train the back-squat in ≥1 training sessions per week. Because of small participant numbers ($n = 6$ for each group), the results of this study may not provide a full representation of the differences between each squat type. Some differences may be due to sampling error.

Before testing, written informed consent was received from each participant, and all testing conditions were examined and approved by the Auckland University of Technology Ethics Committee (14/398).

Procedures

Powerlifters and Olympic Weightlifters. The POW and OLY were required to attend only one session of approximately 3 hours in duration. A full "level 2" anthropometric assessment was performed on all athletes by an experienced International Society for the Advancement of Kinanthropometry anthropometrist, followed by an LBBS 1RM test for the POW, and a HBBS 1RM test for the OLY.

Comparison Group. The recreationally trained athletes were required to attend 4 separate sessions over the course of 1 week: 2 guided 1-hour familiarization sessions, 1 personal familiarization session, and 1 3-hour–long testing session (Figure 2). The first familiarization session comprised of the 1RM testing protocol for HBBS and LBBS with loads up to 60% of self-reported or predicted 1RM. Self-reported 1RM values (performed within the last 6 months) for either back-squat variation were used to estimate load progressions. Pilot testing determined that the load of the unknown back-squat variation would be around 90% of the known back-squat 1RM regardless of which squat style was routinely performed. Thus, the loads for the familiarization session were estimated from 1 known 1RM for 1 back-squat variation and a predicted 1RM at 90% of the known 1RM. The second familiarization session was performed 2 days later and comprised the same HBBS and LBBS protocol in the same order as the first familiarization session, up to 80% 1RM of the self-reported and predicted 1RM for either back-squat variation.

In both the first and second familiarization sessions for each participant, the resistance exercise–specific rating of perceived exertion (RPE) scale (43) was used to ensure that intensity and predicted attempt weight values were correct. In the first familiarization session, an RPE value of 3 or less (i.e., "light to little effort") was expected to be reported in line with the percentages of the 1RM (50 and 60%). If this was not achieved, the predicted weight values were changed for the second familiarization session. In the second familiarization session, the same RPE values of 3 or less were employed for the 50 and 60% of predicted 1RM sets. After that, a self-reported RPE of 5 or less (i.e., "light effort with

Table 3. Distance of COP to bar results.*†

% Range	OLY (mm)	POW (mm)	HBCOM (mm)	LBCOM (mm)
74–83	−19 ± 42	−44 ± 31	−60 ± 45	−57 ± 18
84–93	−20 ± 40	−58 ± 39	−51 ± 42	−72 ± 25
94–99	−23 ± 29	−46 ± 31	−58 ± 35	−59 ± 38
100	−24 ± 40	−74 ± 52	−39 ± 49	−51 ± 18

*COP = center of pressure; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters.
†Negative numbers represent the bar a distance behind the COP. All COP data are presented as mean ± *SD*.

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Table 4. Kinematic results.*†

			HBBS				LBBS			
%1RM Range	Joint	Variable	OLY angle (°)	HBCOM angle (°)	OLY vs. HBCOM Difference; ±90% CI	OLY vs. POW Difference; ±90% CI	POW angle (°)	LBCOM angle (°)	POW vs. LBCOM Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
74–83	Hip	Peak flexion	69 ± 7	64 ± 5	6 ± 7	8 ± 10	59 ± 9	61 ± 4	3 ± 8	3 ± 3
		ROM	100 ± 8	105 ± 9	5 ± 10	6 ± 11	109 ± 11	101 ± 9	9 ± 12	4 ± 4
	Knee	Peak flexion	54 ± 7	59 ± 8	3 ± 9	9 ± 11	62 ± 11	63 ± 8	1 ± 11	4 ± 4
		ROM	116 ± 7	110 ± 11‡	3 ± 11	5 ± 12	114 ± 12	104 ± 10	8 ± 13	5 ± 5
	Ankle	Peak dorsiflexion	90 ± 5	88 ± 6	4 ± 6	2 ± 5	90 ± 5	90 ± 8	0 ± 7	2 ± 4
		ROM	33 ± 4	32 ± 3	0 ± 4	1 ± 6	33 ± 6	30 ± 4	2 ± 6	2 ± 3
84–93	Hip	Peak flexion	69 ± 9	64 ± 6‡	6 ± 8	6 ± 11	59 ± 8	61 ± 3‡	3 ± 7	3 ± 3
		ROM	100 ± 9	105 ± 10	6 ± 11	8 ± 11	111 ± 11	99 ± 9	13 ± 11	5 ± 5
	Knee	Peak flexion	56 ± 7	61 ± 8	4 ± 8	7 ± 11	63 ± 12	67 ± 5	4 ± 10	6 ± 6
		ROM	114 ± 7	107 ± 11	5 ± 10	1 ± 12	113 ± 13	101 ± 6	12 ± 12	6 ± 6
	Ankle	Peak dorsiflexion	91 ± 4	90 ± 6	2 ± 5	2 ± 5	90 ± 5	91 ± 7	1 ± 7	1 ± 3
		ROM	33 ± 4	30 ± 5	2 ± 4	2 ± 6	34 ± 7	30 ± 4	4 ± 6	0 ± 2
94–99	Hip	Peak flexion	71 ± 10	69 ± 6‡	4 ± 9	12 ± 12	59 ± 9	61 ± 5‡	2 ± 9	8 ± 8
		ROM	98 ± 10	100 ± 10	4 ± 11	11 ± 13	110 ± 14	100 ± 10	9 ± 14	0 ± 3
	Knee	Peak flexion	56 ± 7	65 ± 8	8 ± 9	4 ± 10	62 ± 12	68 ± 5	5 ± 10	3 ± 4
		ROM	113 ± 8	103 ± 12	8 ± 12	2 ± 12	114 ± 13	101 ± 7	11 ± 12	6 ± 6
	Ankle	Peak dorsiflexion	90 ± 5	91 ± 6	1 ± 6	0 ± 6	90 ± 5	92 ± 7	2 ± 7	1 ± 2
		ROM	33 ± 4	28 ± 4	4 ± 5	1 ± 6	33 ± 7	29 ± 3	4 ± 6	1 ± 2
100	Hip	Peak flexion	71 ± 9	68 ± 6‡	3 ± 8	12 ± 12	59 ± 10	63 ± 6‡	4 ± 8	5 ± 5
		ROM	97 ± 10	101 ± 10	4 ± 10	11 ± 12	109 ± 13	96 ± 11	13 ± 13	5 ± 5
	Knee	Peak flexion	56 ± 7§	65 ± 6‡§	9 ± 9	7 ± 11	63 ± 12	73 ± 6‡	10 ± 10	7 ± 7
		ROM	113 ± 9	103 ± 9‡	10 ± 10	0 ± 12	113 ± 14‖	95 ± 8‡‖	18 ± 18	8 ± 8
	Ankle	Peak dorsiflexion	90 ± 5	92 ± 6	1 ± 6	0 ± 6	91 ± 6	93 ± 6	3 ± 7	2 ± 2
		ROM	32 ± 3§	27 ± 4§	5 ± 5	1 ± 6	33 ± 8	27 ± 4	6 ± 6	0 ± 4

*1RM = 1 repetition maximum; CI = confidence interval; HBBS = high-bar back-squat; HBCOM = comparison high-bar back-squat; LBBS = low-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; ROM = range-of-motion.

†All angle data are presented as mean ± *SD*.

‡$p \leq 0.05$ HBCOM vs. LBCOM.

§$p \leq 0.05$ OLY vs. HBCOM.

‖$p \leq 0.05$ POW vs. LBCOM.

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Table 5. Kinematic effect sizes and percentage differences.*

			OLY vs. HBCOM		POW vs. LBCOM		OLY vs. POW		HBCOM vs. LBCOM	
% Range	Joint	Variable	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference
74–83	Hip	Peak flexion	0.5‡	4.3	0.2†	2.7	0.4†	16.4	0.7‡	5.1
		ROM	0.3†	7.4	0.4†	8.3	0.3†	8.0	0.6‡	3.9
	Knee	Peak flexion	0.2†	5.1	0.0	1.8	0.4†	12.0	0.6‡	7.1
		ROM	0.2†	2.5	0.3†	8.8	0.2†	1.9	0.7‡	5.2
	Ankle	Peak dorsiflexion	0.3†	1.9	0.0	0.6	0.2†	0.7	0.3†	2.3
		ROM	0.1	15.9	0.2†	10.1	0.1	1.7	0.3‡	5.9
84–93	Hip	Peak flexion	0.4†	7.9	0.2†	3.1	0.3†	16.2	0.8§	4.2
		ROM	0.3†	5.3	0.1	11.9	0.4†	10.2	0.6‡	5.7
	Knee	Peak flexion	0.3†	8.7	0.2†	6.4	0.3†	11.0	0.6‡	9.5
		ROM	0.2†	6.2	0.6‡	12.4	0.1	0.8	0.6‡	6.2
	Ankle	Peak dorsiflexion	0.2†	0.8	0.0	0.8	0.2†	0.4	0.2†	1.2
		ROM	0.2†	10.2	0.3†	14.6	0.2†	2.4	0.0	1.5
94–99	Hip	Peak flexion	0.2†	2.4	0.1	2.9	0.6‡	19.5	2.3§	11.8
		ROM	0.2†	2.3	0.4†	9.4	0.4†	10.9	0.1	0.2
	Knee	Peak flexion	0.4†	13.7	0.2†	7.9	0.2†	9.9	0.3†	4.0
		ROM	0.4†	10.0	0.5‡	12.4	0.1	0.6	0.6‡	1.6
	Ankle	Peak dorsiflexion	0.1	1.0	0.1	1.9	0.0	0.1	0.2†	0.9
		ROM	0.4†	15.4	0.3†	13.9	0.1	1.9	0.2†	3.2
100	Hip	Peak flexion	0.2†	3.8	0.3†	6.6	0.7‡	20.7	1.3§	7.9
		ROM	0.2†	3.6	0.5‡	13.2	0.5‡	10.5	0.6‡	4.9
	Knee	Peak flexion	0.7‡	14.3	0.5‡	13.7	0.3†	10.4	0.9§	10.9
		ROM	0.6‡	9.9	0.8§	18.9	0.0	0.1	0.8§	7.7
	Ankle	Peak dorsiflexion	0.1	1.3	0.2†	2.9	0.0	0.2	0.6‡	1.9
		ROM	0.7‡	18.3	0.5‡	22.0	0.1	1.9	0.0	1.1

*HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; ROM = range-of-motion.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.
§Large effect $d \geq 0.8$.

at-least 6 more repetitions possible") was expected for the 70 and 80% of 1RM sets. If these RPE values were not achieved, the predicted 1RMs for both back-squat variations were changed for the final testing session. In the period between the second familiarization session and the final testing session, a self-directed familiarization session was included for each participant to reinforce the skills learned in the previous familiarization sessions and to provide a chance to practice each bar position before the testing. Each participant was asked not to exceed an RPE of 5 in this session and to do no more than 3 sets. The final testing session was performed 3 days later and comprised a full anthropometric assessment, followed by a 1RM test of both the HBBS and LBBS in random order, so that half of the comparison group performed the HBBS first and the other half performed the LBBS first. This randomized order was employed to minimize any fatigue affect from performing 2 maximal squat tests in one testing session.

Back-Squat 1 Repetition Maximum Testing Protocol

All squats were completed in line with the International Powerlifting Federation's competition rules (21). Both the HBBS and LBBS were deemed to be successful lifts if the athlete was able to safely lower the bar to a minimum accepted depth (the top surface of the legs at the hip joint is lower than the top of the knees) or lower, through a bending of the knees, and then recover at will to a stance with knees locked, without the aid of any spotters. The OLY participants were instructed to squat to the usual depth they perform in training. Specific focus was placed on ensuring correct depth was obtained, the legs were

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Table 6. Kinetic results 74–83% 1RM.*†

		HBBS				LBBS			
Phase	Variable	OLY	HBCOM	OLY vs. HBCOM Difference; ±90% CI	OLY vs. POW Difference; ±90% CI	POW	LBCOM	POW vs. LBCOM Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
Eccentric	Mean bar v ($m \cdot s^{-1}$)	0.51 ± 0.13	0.44 ± 0.11	0.09 ± 0.19	0.05 ± 0.10	0.54 ± 0.12‡	0.38 ± 0.09‡	0.20 ± 0.20	0.06 ± 0.07
	Peak Fv ($N \cdot kg^{-1}$)	38 ± 3§	26 ± 4§	10 ± 10	1 ± 3	37 ± 2.69‡	26 ± 3‡	9 ± 9	1 ± 2
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,746 ± 1,080§	845 ± 318§	2,190 ± 2,190	862 ± 948	2,294 ± 824	1,102 ± 339	1,213 ± 1,213	231 ± 319
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,657 ± 1,788‖	1,570 ± 539	2,396 ± 2,396	1,319 ± 1,553	3,058 ± 1,376‖	1,877 ± 415	1,377 ± 1,641	337 ± 436
Concentric	Mean bar v ($m \cdot s^{-1}$)	0.51 ± 0.05	0.49 ± 0.11	0.07 ± 0.08	0.03 ± 0.06	0.57 ± 0.08‡	0.55 ± 0.11‡	0.09 ± 0.10	0.04 ± 0.05
	Peak Fv ($N \cdot kg^{-1}$)	38 ± 3§	31 ± 27§	10 ± 10	1 ± 3	37 ± 3‡	27 ± 4‡	8 ± 8	1 ± 1
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,013 ± 737	816 ± 416	1,131 ± 1,131	311 ± 1,046	2,002 ± 1,089	707 ± 166	1,319 ± 1,317	85 ± 283
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3110 ± 1502	1391 ± 608	1695 ± 1922	344 ± 1634	3287 ± 1474	1258 ± 328	2084 ± 2084	98 ± 359

*1RM = 1 repetition maximum; CI = confidence interval; Fv = vertical force; HBBS = high-bar back-squat; HBCOM = comparison high-bar back-squat; LBBS = low-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.

†All kinetic data are presented as mean ± *SD*.

‡$p \leq 0.05$ POW vs. LBCOM.

§$p \leq 0.05$ OLY vs. HBCOM.

‖$p \leq 0.05$ OLY vs. POW.

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TABLE 7. Kinetic effect sizes and percentage differences 74–83% 1RM.*

		OLY vs. HBCOM		POW vs. LBCOM		OLY vs. POW		HBCOM vs. LBCOM	
Phase	Variable	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference
Eccentric	Mean bar v ($m \cdot s^{-1}$)	0.3†	15.9	1.0§	40.2	0.3†	5.3	0.7‡	12.7
	Peak Fv ($N \cdot kg^{-1}$)	1.5§	43.7	1.9§	40.9	0.2†	1.3	0.4†	0.7
	RFD (0–50 ms) ($N \cdot s^{-1}$)	1.2§	224.8	0.9§	108.1	0.6‡	19.7	0.7‡	23.3
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.8§	132.9	0.6‡	62.9	0.6‡	19.6	0.7‡	16.4
Concentric	Mean bar v ($m \cdot s^{-1}$)	0.6‡	2.4	0.6‡	4.0	0.4†	11.9	0.8§	11.7
	Peak Fv ($N \cdot kg^{-1}$)	1.6§	42.0	1.6§	37.6	0.2†	0.7	1.1§	2.5
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.8§	146.7	0.7‡	183.3	0.2†	0.6	0.3†	15.5
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.6‡	123.6	0.8§	161.2	0.1	5.4	0.2†	10.5

*1RM = 1 repetition maximum; Fv = vertical force; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.
§Large effect $d \geq 0.8$.

completely locked out at the conclusion of each repetition, and no downward movement was observed on the ascent.

Before testing, each participant's beltless 1RM was estimated. If in normal training, the participant did not use a weight belt, the athlete's predicted beltless 1RM was used. If the participant used a weight belt in normal training, and had a known belted 1RM, this belted 1RM was used to predict the athlete's beltless 1RM. Pilot testing determined that the beltless 1RM is approximately 90% of a belted 1RM. Weightlifting shoes (comprised of a hard sole and a slightly raised heel) were required to be worn by all participants, and the heel height was required to be within the range of 1.5–2.0 cm. All participants were accustomed to wearing weightlifting shoes. No other supportive aids beyond the use of wrist wraps were allowed to be worn during the test. Before all testing procedures, each participant completed a standardized dynamic warm-up.

The 1RM testing protocol was adapted from Matuszak et al. (26) and consisted of the participants performing 8 repetitions at 50% of the predicted 1RM, 3 repetitions at 60%, 2 repetitions at 70%, and 1 repetition at 80 and 90%. Additional warm-up sets, before the initial 8 repetition set with 50% 1RM, were permitted with <50% 1RM load if the participant desired to do so as to better replicate their normal warm-up procedures. After the 90% of predicted 1RM lift, the participant was consulted as to what weight they would like to attempt for a maximal 1RM lift. An experienced strength coach along with the use of a Gymaware Powertool (Kinetic Performance Technology, Canberra, Australia) to measure the mean concentric velocity of the movement assisted athletes in attempt selection to get as close to a true beltless 1RM as possible. Previous research has shown that maximal squat attempts performed by experienced lifters are typically performed at approximately 0.2 $m \cdot s^{-1}$ (0.24 ± 0.04 $m \cdot s^{-1}$) (43). Commonly, a lift at 95% 1RM was performed before attempting the predicted maximal 1RM. After each successful attempt, small weight increments (1–5 kg) were made to obtain a true maximum. Between 3 and 5 minutes of rest was allowed between sets before the next weight was attempted.

Biomechanical Instrumentation

Two embedded force platforms (Model AM6501; Bertec Corp., Columbus, OH, USA), were used to collect all kinetic squat data at a sampling rate of 1,000 Hz. The kinetic variables of interest included mean bar velocity ($m \cdot s^{-1}$); peak Fv ($N \cdot kg^{-1}$); rate of force development (RFD) (0–50 ms) ($N \cdot s^{-1}$); and RFD (0–100 ms) ($N \cdot s^{-1}$) for both the eccentric and concentric phases. The RFD variables were chosen in line with previous squat research (8). Mean bar velocity was chosen over peak bar velocity for a better representation of each athlete's ability to move load throughout the whole lifting phase (concentric/eccentric) (22). Rate of force development is the change in force over a given time (33), and the eccentric phase of each movement is where the body lowers and slows to

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Table 8. Kinetic results 84–93% 1RM.*†

		HBBS				LBBSt			
Phase	Variable	OLY	HBCOM	OLY vs. HBCOM Difference; ±90% CI	OLY vs. POW Difference; ±90% CI	POW	LBCOM	POW vs. LBCOM Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
Eccentric	Mean bar *v* ($m \cdot s^{-1}$)	0.48 ± 0.09	0.39 ± 0.08	0.09 ± 0.19	0.00 ± 0.10	0.51 ± 0.10‡	0.35 ± 0.10‡	0.16 ± 0.16	0.04 ± 0.04
	Peak *Fv* ($N \cdot kg^{-1}$)	40 ± 3§	28 ± 5 §	10 ± 10	2 ± 3	38 ± 3‡	27 ± 3‡	10 ± 10	0 ± 3
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,258 ± 943	857 ± 737	1,088 ± 1,188	517 ± 957	1,857 ± 648‡	493 ± 112‡	1,425 ± 1,425	362 ± 745
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,413 ± 1,587	1,552 ± 1,233	1,727 ± 2,147	715 ± 1,648	2,896 ± 1,226‡	950 ± 74‡	1,987 ± 1,987	602 ± 1,247
Concentric	Mean bar *v* ($m \cdot s^{-1}$)	0.41 ± 0.06	0.40 ± 0.05	0.06 ± 0.06	0.00 ± 0.07	0.44 ± 0.09‡	0.42 ± 0.09‡	0.06 ± 0.08	0.03 ± 0.05
	Peak *Fv* ($N \cdot kg^{-1}$)	39 ± 4§	28 ± 5§	10 ± 10	2 ± 4	38 ± 3‡	29 ± 4‡	8 ± 8	1 ± 3
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,278 ± 921	889 ± 324	1,282 ± 1,282	865 ± 1,023	1,617 ± 838	705 ± 243	1,036 ± 1,036	183 ± 333
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,303 ± 1,632	1,325 ± 674	1,930 ± 1,994	1,024 ± 1,727	2,686 ± 1,448‡	964 ± 223‡	1,871 ± 1,871	357 ± 719

*1RM = 1 repetition maximum; CI = confidence interval; *Fv* = vertical force; HBBS = high-bar back-squat; HBCOM = comparison high-bar back-squat; LBBS = low-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.

†All kinetic are data presented as mean ± *SD*.

‡$p \leq 0.05$ POW vs. LBCOM.

§$p \leq 0.05$ OLY vs. HBCOM.

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Table 9. Kinetic effect sizes and percentage differences 84–93% 1RM.*

		OLY vs. HBCOM		POW vs. LBCOM		OLY vs. POW		HBCOM vs. LBCOM	
Phase	Variable	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference
Eccentric	Mean bar v ($m \cdot s^{-1}$)	0.3†	22.7	0.9§	46.0	0.0	6.1	0.9§	10.5
	Peak Fv ($N \cdot kg^{-1}$)	1.5§	42.5	2.1§	39.6	0.4†	4.3	0.1	2.1
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.7‡	163.5	1.5§	276.5	0.4†	21.6	0.4†	73.7
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.6‡	119.9	1.1§	204.8	0.3†	17.8	0.4†	63.3
Concentric	Mean bar v ($m \cdot s^{-1}$)	0.7‡	5.5	0.5‡	5.1	0.0	6.8	0.4†	7.7
	Peak Fv ($N \cdot kg^{-1}$)	1.3§	39.7	1.6§	32.0	0.4†	3.6	0.3†	2.1
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.8§	156.3	0.9§	129.2	0.6‡	40.9	0.5‡	26.0
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.7‡	149.3	0.9§	178.6	0.4†	23.0	0.4†	37.4

*1RM = 1 repetition maximum; Fv = vertical force; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.
§Large effect $d \geq 0.8$.

a point of zero velocity, immediately before the start of the concentric ascent. The eccentric RFD is measured in the time before this change from the eccentric phase to the concentric phase. The 2 force platforms were arranged next to each other in the middle of the collection space to increase the chances of obtaining complete foot contact from each foot during the required movements. Kinematics were collected by 9 infrared cameras (T10S; Vicon Motion System Ltd., Oxford, United Kingdom) strategically placed around the force platforms in the collection space. The cameras were arranged so that each marker was always visible to a minimum of 3 cameras to allow for reconstruction of three-dimensional (3D) trajectories. The collection space was calibrated with an error of no greater than 0.2 (route mean squared in camera pixels; the difference between the two-dimensional image of each marker on the camera sensor and the 3D reconstructions of those markers projected back to the cameras sensor) for each camera before each data collection session, and a point of origin was positioned at the corner of one of the force platforms to establish a local relationship between the camera positions and the laboratory origin. Data from 8 reflective markers (10-mm diameter) placed in specific locations were used to analyze bar path and joint angles throughout the squat movement using Vicon Nexus software (Version 1.8.5; Vicon Motion System Ltd., Oxford, United Kingdom). The joint angles were calculated as the angle between a parent segment (i.e., thigh or femur) and a child segment (i.e., shank or tibia). Markers were placed in the center of both ends of the barbell and on the right side of the athletes' bodies in specific anatomical locations after previous research (28) (Figure 1). The markers were placed on the following locations: acromion process, greater trochanter, lateral epicondyle of the femur, lateral malleolus, top of the heal lift of the lifting shoe and in line with the lateral malleolus, and base of the fifth metatarsal to create 5 rigid segments.

Data Reduction

Subsequent to the testing sessions, the 2 force platforms were combined and all data were filtered with a low-pass fourth-order zero-lag Butterworth filter using a cut-off frequency of 16 Hz in a custom-made LabVIEW program (Version 14.0; National Instruments Corp., Austin, TX, USA) based on residual analysis and visual inspection of the kinematic and kinetic data. Kinematic variables of interest were gathered through an individual analysis within the start and finish of the squat to calculate the range-of-motion (ROM) (peak flexion–initial or finishing flexion) and peak flexion angles for the hip, knee, and ankle joints. Peak joint flexion was recorded as the angle at the lowest point of the lift, and peak extension at the highest point of the lift. The hip ROM in the sagittal plane was derived from the anterior angle between the thorax (trunk) and the thigh, the knee ROM was derived from the posterior angle between the thigh and the shank, and the ankle ROM

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Table 10. Kinetic results 94–99% 1RM.*†

		HBBS				LBBS			
Phase	Variable	OLY	HBCOM	OLY vs. HBCOM Difference; ±90% CI	OLY vs. POW Difference; ±90% CI	POW	LBCOM	POW vs. LBCOM Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
Eccentric	Mean bar *v* ($m \cdot s^{-1}$)	0.47 ± 0.09	0.36 ± 0.10	0.12 ± 0.15	0.04 ± 0.13	0.45 ± 0.12	0.34 ± 0.07	0.10 ± 0.14	0.03 ± 0.07
	Peak *Fv* ($N \cdot kg^{-1}$)	41 ± 4‡	29 ± 4‡§	11 ± 11	2 ± 4	39 ± 3‖	28 ± 3§‖	11 ± 11	1 ± 1
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,018 ± 1,110	811 ± 500	1,272 ± 1,479	383 ± 1,275	1,618 ± 1,107	687 ± 140	893 ± 1,207	123 ± 504
	RFD (0–100 ms) ($N \cdot s^{-1}$)	2,953 ± 1,658	1,413 ± 957	1,344 ± 2,274	477 ± 1,665	2,371 ± 1,266	1,071 ± 402	1,302 ± 1,418	300 ± 848
Concentric	Mean bar *v* ($m \cdot s^{-1}$)	0.32 ± 0.03	0.31 ± 0.05	0.06 ± 0.06	0.02 ± 0.05	0.31 ± 0.05	0.31 ± 0.04	0.01 ± 0.06	0.02 ± 0.03
	Peak *Fv* ($N \cdot kg^{-1}$)	41 ± 5‡	29 ± 4‡	11 ± 11	2 ± 4	39 ± 3‖	30 ± 4‖	9 ± 9	1 ± 1
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,083 ± 906‡	706 ± 525‡	992 ± 1,154	327 ± 935	1,595 ± 818	575 ± 342	1,016 ± 1,016	141 ± 294
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,425 ± 1,412	1,062 ± 815	1,880 ± 1,880	498 ± 1,481	2,761 ± 1,258‖	870 ± 461‖	1,866 ± 1,866	224 ± 542

*1RM = 1 repetition maximum; CI = confidence interval; *Fv* = vertical force; HBBS = high-bar back-squat; HBCOM = comparison high-bar back-squat; LBBS = low-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.
†All kinetic data are presented as mean ± *SD*.
‡$p \leq 0.05$ OLY vs. HBCOM.
§$p \leq 0.05$ HBCOM vs. LBCOM.
‖$p \leq 0.05$ POW vs. LBCOM.

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Table 11. Kinetic effect sizes and percentage differences 94–99% 1RM.*

		OLY vs. HBCOM		POW vs. LBCOM		OLY vs. POW		HBCOM vs. LBCOM	
Phase	Variable	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference
Eccentric	Mean bar v ($m \cdot s^{-1}$)	0.6‡	27.9	0.5‡	33.5	0.2†	2.4	0.4†	6.4
	Peak Fv ($N \cdot kg^{-1}$)	1.2§	43.9	2.0§	41.5	0.3†	4.2	1.3§	2.4
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.6‡	148.9	0.5‡	135.6	0.2†	24.7	0.2†	18.0
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.4†	109.0	0.7‡	121.4	0.2†	24.5	0.3†	31.9
Concentric	Mean bar v ($m \cdot s^{-1}$)	1.2§	5.5	0.1	1.1	0.3†	3.9	0.5‡	2.7
	Peak Fv ($N \cdot kg^{-1}$)	1.2§	42.4	1.6§	32.1	0.3†	4.1	0.7‡	3.5
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.6‡	195.0	0.8§	177.6	0.6‡	30.6	0.4†	22.9
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.7‡	222.3	0.9§	217.2	0.2†	24.0	0.4†	22.1

*1RM = 1 repetition maximum; Fv = vertical force; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.
§Large effect $d \geq 0.8$.

was derived from the angle between the shank and the foot. In all cases, the actual angle is presented as opposed to the raw angle (Figure 1). To obtain kinetic variables of interest, all repetitions were individually analyzed during the eccentric phase (from the initiation of a negative [downward] velocity of the right-side bar marker to the instant the marker reached zero velocity [full depth]) and concentric phase (from the initiation of a positive [upward] velocity of the right-side bar marker to the instant the marker reached zero velocity a second time [the top]).

To obtain kinematic variables of interest, all repetitions were individually analyzed within the start and finish of the squat movement to calculate the ROM (peak flexion–initial flexion) and peak flexion angles for the hip, knee, and ankle joints. From the sagittal plane, the hip ROM was derived from the anterior angle between the thorax (trunk) and the thigh, the knee ROM was derived from the posterior angle between the thigh and the shank and the ankle ROM was derived from the angle between the shank and the foot. In all cases, the actual angle is presented as opposed to the raw (Figure 1).

Statistical Analyses

Before analyses, data were split into 4 categories according to trials where a single squat was completed and the %1RM load achieved in testing: (a) 74–83%, (b) 84–93%, (c) 94–99%, and (d) 100%. This was necessary because of the variation in the number of single repetition trials completed before a true 1RM was achieved between participants. If multiple trials were completed within a 1RM range for a participant, the results were averaged so each participant effectively had 1 trail per category. Generalized linear mixed models using a normal distribution with an identity link and unstructured covariance structure were used to estimate the difference in outcome variables between bar height and subject group across all 4 load groups, while adjusting for the random effect of subject. In an unstructured covariance matrix, each variance and each covariance value is estimated uniquely from the data, resulting in the best possible model fit (39). Robust standard errors, constructed using the "sandwich estimator" of the covariance structure, were used to control for possible misspecifications of the correlation structure. An alpha of 0.05 was used to determine significant associations. Multiple pairwise comparisons were corrected for inflation of type 1 error using the Bonferroni method (e.g., for all pairwise comparisons in a fixed factor with 3 groups, significance level was divided by 3). For all variables, Cohen's d statistic was calculated as the estimated marginal means divided by the square root of N multiplied by the SE (i.e., the SD) to provide additional information on the magnitude of the associations, with 0.2, 0.5, and 0.8 representing small, moderate, and large effects, respectively (3). The analysis used IBM SPSS Statistics v. 23.0.0.0 (IBM, Armonk, NY, USA) software.

Results

Initially, a comparison of the HBBS performed by the OLY and comparison groups (comparison high-bar back-squat

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Table 12. Kinetic results 100% 1RM.*†

		HBBS				LBBS			
Phase	Variable	OLY	HBCOM	OLY vs. HBCOM Difference; ±90% CI	OLY vs. POW Difference; ±90% CI	POW	LBCOM	POW vs. LBCOM Difference; ±90% CI	HBCOM vs. LBCOM Difference; ±90% CI
Eccentric	Mean bar *v* ($m \cdot s^{-1}$)	0.48 ± 0.09‡	0.34 ± 0.09‡	0.14 ± 0.14	0.03 ± 0.11	0.44 ± 0.14§	0.31 ± 0.06§	0.14 ± 0.14	0.04 ± 0.07
	Peak *Fv* ($N \cdot kg^{-1}$)	42 ± 4‡	29 ± 4‡	13 ± 13	2 ± 3	40 ± 2§	29 ± 3§	11 ± 11	0 ± 2
	RFD (0–50 ms) ($N \cdot s^{-1}$)	2,240 ± 852‡	634 ± 372‡	1,606 ± 1,606	490 ± 905	1,750 ± 878§	375 ± 337§	1,375 ± 1,375	258 ± 413
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,062 ± 1,681	1,052 ± 650	2,010 ± 2,010	406 ± 1,660	2,656 ± 1,485§	676 ± 581§	1,980 ± 1,980	376 ± 769
Concentric	Mean bar *v* ($m \cdot s^{-1}$)	0.22 ± 0.03	0.20 ± 0.03	0.02 ± 0.04	0.01 ± 0.05	0.21 ± 0.06	0.23 ± 0.05	0.03 ± 0.07	0.04 ± 0.06
	Peak *Fv* ($N \cdot kg^{-1}$)	41 ± 4‡	30 ± 3‡	12 ± 12	2 ± 4	40 ± 2§	31 ± 3§	9.00 ± 9.00	1 ± 2
	RFD (0–50 ms) ($N \cdot s^{-1}$)	1,734 ± 916‡	629 ± 248‡	1,105 ± 1,105	86 ± 1,197	1,820 ± 1,332	507 ± 222	1,313 ± 1,313	122 ± 179
	RFD (0–100 ms) ($N \cdot s^{-1}$)	3,218 ± 1,572‡	1,049 ± 480‡	2,169 ± 2,169	202 ± 1,972	3,016 ± 2,153	676 ± 254	2,341 ± 2,341	374 ± 528

*1RM = 1 repetition maximum; CI = confidence interval; *Fv* = vertical force; HBBS = high-bar back-squat; HBCOM = comparison high-bar back-squat; LBBS = low-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.

†All kinetic data are presented as mean ± *SD*.

‡$p \leq 0.05$ OLY vs. HBCOM.

§$p \leq 0.05$ POW vs. LBCOM.

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Table 13. Kinetic effect sizes and percentage differences 100% 1RM.*

		OLY vs. HBCOM		POW vs. LBCOM		OLY vs. POW		HBCOM vs. LBCOM	
Phase	Variable	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference	Effect size	% Difference
Eccentric	Mean bar v ($m \cdot s^{-1}$)	0.9§	40.0	0.9§	46.4	0.2†	7.7	0.5‡	11.2
	Peak Fv ($N \cdot kg^{-1}$)	1.9§	44.8	2.9§	39.7	0.3†	3.5	0.2†	0.2
	RFD (0–50 ms) ($N \cdot s^{-1}$)	1.3§	253.5	1.1§	366.4	0.4†	28.0	0.5‡	68.9
	RFD (0–100 ms) ($N \cdot s^{-1}$)	0.8§	191.1	0.9§	292.9	0.2†	15.3	0.4†	55.6
Concentric	Mean bar v ($m \cdot s^{-1}$)	0.4†	9.9	0.3†	10.6	0.1	4.5	0.5‡	17.7
	Peak Fv ($N \cdot kg^{-1}$)	1.7§	39.3	2.1§	29.2	0.3†	3.2	0.7‡	4.5
	RFD (0–50 ms) ($N \cdot s^{-1}$)	0.9§	175.6	0.7‡	259.1	0.0	4.7	0.6‡	24.1
	RFD (0–100 ms) ($N \cdot s^{-1}$)	1.0§	206.7	0.8§	346.5	0.1	6.7	0.6‡	55.3

*1RM = 1 repetition maximum; Fv = vertical force; HBCOM = comparison high-bar back-squat; LBCOM = comparison low-bar back-squat; OLY = Olympic weightlifters; POW = powerlifters; RFD = rate of force development.
†Small effect $d \geq 0.2$.
‡Moderate effect $d \geq 0.5$.
§Large effect $d \geq 0.8$.

[HBCOM]) and the LBBS performed by the POW and comparison groups (comparison low-bar back-squat [LBCOM]) was completed to determine whether the comparison group data could be combined with the OLY or the POW for the high- and low-bar positions, respectively. Significant joint angle differences were observed in knee flexion ($p = 0.04$) and ankle ROM ($p = 0.04$) at 100% of 1RM for HBBS (OLY vs. HBCOM) and in knee ROM ($p = 0.02$) at 100% 1RM for the LBBS (POW vs. LBCOM). Significant differences for several kinetic variables across all 4 percentage ranges of 1RM for both HBBS (OLY vs. HBCOM) and LBBS (POW vs. LBCOM) were also observed. Therefore, in the following sections, the data have been analyzed with all 4 groups displayed independently.

Load

The mean loads are presented in Tables 1 and 2. No significant differences were observed between OLY and POW and between HBCOM and LBCOM. However, on an average, the POW group lifted greater loads compared with the OLY group across all ranges of load ($d = 0.3$, 0.2, 0.2, and 0.2 for ranges of 74–83%, 84–93%, 94–99%, and 100% 1RM, respectively). Small effect sizes indicated that greater loads and loads relative to body weight were lifted by the LBCOM group than the HBCOM group for the 74–83% ($d = 0.3$ and 0.3, respectively), and 84–93% ($d = 0.3$ and 0.4, respectively) 1RM ranges, but only for load at 100% 1RM ($d = 0.4$). Moderate effect sizes indicated that greater loads were lifted by the LBCOM in comparison with the HBCOM group at 94–99% 1RM in both load and load relative to body weight ($d = 0.5$ and 0.6, respectively) and at 100% 1RM in load relative to body weight ($d = 0.5$).

Center of Pressure

The mean distances of the bar from the center of pressure (COP) are presented in Table 3. In the experienced OLY and POW groups, there is a distinct difference between the 2 bar positions. The LBBS performed by the POW shows a greater average distance from the bar to the COP. In the less experienced COM group, the same difference is generally observed between the HBBS and LBBS, but is much less pronounced.

Kinematics

Differences in the estimated marginal means for the kinematic variables are presented in Tables 4 and 5. No significant differences were observed between the OLY and POW groups in any condition. A significantly larger knee flexion angle was observed in the HBCOM when compared with the OLY group ($p = 0.04$; $d = 0.7$; % Difference = 14.3) at 100% 1RM. Conversely, the OLY group displayed a significantly larger ankle ROM than the HBCOM group at 100% ($p = 0.04$; $d = 0.07$; % Difference = 18.3). The only significant difference between the POW and LBCOM groups was observed at 100% 1RM, with the POW group demonstrating a significantly larger knee ROM ($p = 0.02$; $d = 0.8$; % Difference = 18.9). The majority of significant results were observed between the HBCOM and

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LBCOM. Significant differences were observed in knee ROM at 74–83% 1RM ($p = 0.04$), peak hip flexion at 84–93% 1RM ($p = 0.02$), peak hip flexion at 94–99% 1RM ($p = 0.00$) and peak hip flexion ($p = 0.00$), peak knee flexion ($p = 0.01$), and knee ROM at 100% 1RM ($p = 0.02$). In all cases, the HBCOM group displayed larger angles, except for peak knee flexion at 100% 1RM where the LBCOM was greater. No significant interactions between load and group were detected.

Kinetics

Kinetic differences in estimated marginal means are presented in Tables 6–13. The only significant difference observed between the OLY and POW groups across all percentage ranges of 1RM was in the eccentric phase RFD (0–50 ms) at 74–83% 1RM ($p = 0.03$). Small effects were observed for a variety of variables across all 4 ranges of load (%1RM). Moderate kinetic effects showing a greater OLY RFD were also observed in the eccentric phase of the squat at 74–83% 1RM 0–50 milliseconds ($d = 0.6$) and 0–100 milliseconds ($d = 0.6$). Moderately larger effects were also observed in the concentric phase in the OLY at 84–93% 1RM at 0–50 milliseconds ($d = 0.6$) and at 94–99% 1RM (0–50 ms) ($d = 0.6$). Only 1 significant difference between the HBCOM and LBCOM was observed. The HBCOM group produced a significantly greater peak *Fv* in the eccentric phase at 94–99% 1RM ($p = 0.05$; $d = 0.9$; % Difference = 2.4) (Tables 10 and 11). A large number of significant differences ($p \leq 0.05$) were observed across all load ranges, in both the eccentric and concentric phases for OLY vs. HBCOM and for POW vs. LBCOM (Tables 6, 8, 10, 12). In all cases of significant difference, the more experienced OLY and POW groups produced larger forces than those produced by the less experienced HBCOM and LBCOM groups, respectively.

Discussion

The purpose of this study was to compare and contrast the differences in kinematics and kinetics between the HBBS and LBBS to understand why the LBBS might typically allow for greater loads to be lifted (32). Originally, the HBBS and LBBS were compared by combining experienced populations (OLY and POW) with the same bar position in resistance-trained individuals (HBCOM and LBCOM). However, initial analyses revealed differences between groups using the same bar position (i.e., between HBCOM and OLY and between LBCOM and POW, respectively). Therefore, each group was compared independently to examine the kinematic and kinetic differences that arise as a function of bar position (i.e., high-bar and low-bar position) and experience level (i.e., OLY high-bar vs. POW low-bar).

To the best of our knowledge, this is the first study to compare the kinematic and kinetic differences of the HBBS and LBBS using loads ≥90% 1RM. The main findings of this investigation were: (a) statistically significant results were observed in both joint angles and kinetics between the OLY and HBCOM groups and between POW and LBCOM groups; (b) although not significant, a small effect size indicated that greater loads were lifted for each of the percentage 1RM ranges for the LBBS when comparing the POW vs. OLY ($d = 0.2$–0.3). In addition, small ($d \geq 0.2$) and moderate ($d \geq 0.5$) effect sizes indicated that the LBCOM group lifted greater loads and loads relative to body weight across all ranges of %1RM; (c) no significant differences were observed in kinematics between the OLY and POW groups, in any conditions, and only one significant difference was observed between the OLY and POW groups in kinetics. However, small ($d \geq 0.2$), moderate ($d \geq 0.5$), and large ($d \geq 0.8$) effects were observed across all ranges of load between OLY and POW; (d) significantly larger joint angles were observed on the HBCOM in comparison with the LBCOM in knee ROM at 74–83% and 100% 1RM and peak flexion at 84–93%, 94–99%, and 100% 1RM. The LBCOM, however, did produce a larger knee flexion angle at 100% 1RM than the HBCOM; and (e) only 1 significant difference was observed between the HBCOM and LBCOM groups in kinetics. The HBCOM group produced a significantly larger peak *Fv* at 94–99% 1RM in the eccentric phase.

Surprisingly, no significant differences were observed between the experienced OLY and POW groups for any joint angles. It was expected that the OLY would display a greater angle at peak hip flexion because of the more upright torso position and a smaller knee flexion angle. In this study, small to moderate magnitudes of effect ($d \geq 0.2$–0.5) were observed at all 4 percentages of 1RM, indicating that the OLY group demonstrated a larger hip angle displayed at peak flexion by the OLY group at all percentages of 1RM tested. Previous research by Fry et al. (14) and Wretenberg et al. (41) demonstrated a larger hip angle in the HBBS and a greater forward lean in the LBBS. However, the squats were performed only at 50% and 65% 1RM, respectively, in these aforementioned studies, and the results also failed to reach statistical significance. Therefore, it is possible to surmise that OLY consistently demonstrate a larger hip angle and therefore a more upright torso position when performing the HBBS when compared with the LBBS performed by POW. The knee joint findings of this study were similar to those reported in other studies (5,11,13,17,18,20,24,27,37,38), and it appears that the OLY displays a smaller peak knee flexion angle (i.e., greater depth) than what is seen during the POW. However, the difference was not pronounced, as there were no significant differences observed; but, there were small to moderate magnitudes of change ($d \geq 0.2$–0.5).

Interestingly, however, significant differences were observed in the hip and knee joints, between the HBBS performed by the HBCOM group, and the LBBS performed by the LBCOM. The significant differences between these

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2 groups in joint angles are in line with the previous literature, and this indicates that there may have been an influence of experience on the significant results in this study and in the findings of previous research. The smaller hip angle, and greater knee angle shown by the POW group in this study, indicates a greater posterior displacement of the hip, a more vertical shank, and therefore a greater ankle angle. However, this study showed no significant differences in ankle joint angles between the OLY and POW groups. Instead, only 1 significant difference was presented in the ankle ROM between OLY and HBCOM at 100% 1RM ($p = 0.04$; $d = 0.7$; % Difference = 18.3). Previous investigations have shown no definitive differences between the ankle joint angles of the HBBS and LBBS (13,17,24,34,37). The ankle joint angle results of this study further support these previous findings between experienced populations (i.e., OLY and POW), but may indicate differences in experienced vs. less experienced groups of HBBS practitioners (i.e., OLY and HBCOM) at maximal effort.

The upper body has a larger mass than the lower body, and therefore humans are inherently unstable and require effective control mechanisms to constantly resist perturbation (40). This inherent instability is expressed in 3 planes of motion when load is added to the upper body via a barbell, as in the case of the HBBS and LBBS (35). The COP is the point on the ground at which the *Fv* vector originates and is a representation of the center of mass (COM) that accounts for the whole body's weight (including the external bar load) (1). It can be argued that the COM/COP will be in the same position with both the HBBS and LBBS; however, the variation in position of the bar forces the segments of the body to adapt differently to maintain the COM within the athlete's BOS, and therefore combat a loss of balance. A change in 1 body segment will typically result in a change in the other segments (12). The distance of the bar from the COP can help indicate the level of change in these segments, particularly when paired with kinematic joint angle data. The results of this study indicate that the mechanisms the body employs to maintain the balance of its system are concentrated at the hip and not at the knee or ankle joint. At the deepest part of each squat, we found the distance of the bar behind the COP was larger in the LBBS (55 ± 39 mm) than in the HBBS (21 ± 36 mm) (Table 3). Anthropometric differences (e.g., lower limb length) between participants here would create variability if such a measure was to come from a joint center. Instead, the distance from the COP accounts better for the combined mass of the participant and external bar load. These findings exemplify the effects of the low-bar position being further down the back on the lower trapezius musculature and also indicate a more vertical torso in the HBBS. To maintain the position of the barbell on the shoulders and to keep the body's COM within the BOS, the lifter must adopt a smaller torso angle when performing the LBBS. In addition, a wider stance is also often employed when performing the LBBS (10), and anecdotally it is performed to suit the hip structure of the lifter to allow them to obtain the required depth. An increased stance width also acts to effectively increase the BOS, and therefore allows for the bar to be a further distance from the COP, without exiting the BOS. Thus, the smaller hip angle demonstrated in this study may allow greater loads to be lifted with the LBBS, because of the decreased moment arm, greater emphasis on the strong hip musculature, and the aforementioned increased stability (34,37).

The only significant difference observed between the OLY and POW groups across all percentage ranges of 1RM was in the eccentric phase RFD (0–50 ms) at 74–83% 1RM ($p = 0.03$). However, small ($d \geq 0.2$) and moderate ($d \geq 0.5$) magnitudes of change were observed for several variables (Tables 7, 9, 11, 13). The OLY and POW who took part in this study were all of a high level, and consequently they lifted loads that were similar to each other when presented relative to body weight, but not in terms of actual load (Tables 1 and 2). Although not statistically significant, the POW on an average lifted greater loads for each percentage of 1RM. Previous research has shown that as load is increased, there is a resulting increase in the *Fv* produced that is proportionate to the increase in load (6,7,13,23,42). With this in mind, it was expected that the results of this study would show that the POW had the ability to generate greater *Fv* levels during the LBBS, because of the larger loads typically lifted. However, this did not occur. Instead, no significant differences were observed between the POW and OLY groups, and only small effects ($d \geq 0.2$) were observed for *Fv*. These effects are also in direct contrast to Goodin (15), who showed the HBBS to produce larger *Fv*, when compared with the LBBS, with loads of 20–80% 1RM, in HBBS-dominant athletes. In this investigation, the *Fv* levels were shown to be significantly greater in the LBBS than in the HBBS only between the less experienced HBCOM and LBCOM groups in the eccentric phase at 94–99% 1RM ($p = 0.05$; $d = 1.3$; % Difference = 2.4). This indicates that the LBBS may in fact be a more efficient technique of squatting large loads in proportion to the lifter's bodyweight. Even though greater loads were lifted by the POW, when compared with the OLY for each set, the *Fv* produced was relatively the same; thus, the mechanical advantage can be attributed to kinematic joint angle differences. An analysis of the lower limb and trunk muscle activity throughout the squat for both the HBBS and LBBS is necessary to supplement these conclusions. Such an analysis will create a greater understanding as to the level of muscle mass that is deemed to be active throughout each squat style. These findings may provide an insight into the reasons for differing kinetic results, through muscle activity results.

The resistance-trained men in this study were recruited as a comparison group, and they did not have any specific expertise in either the HBBS or LBBS. As a result, the techniques displayed by the comparison group had many significant kinetic differences when compared with the

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well-trained OLY and POW athletes (Tables 6–13). In addition, significant differences were also observed in several joint angles between the OLY and POW groups versus the HBCOM and LBCOM groups (Tables 4 and 5). Therefore, it can be concluded that resistance training experience and technical proficiency have a strong influence on the associated joint angle kinematics and kinetics. Thus, the level of experience of an individual may be a useful predictor of squatting technical performance. This notion and the results of this study are supported by the work of Miletello et al. (30), who reported differences in kinetic and kinematic variables measured at the knee when 3 different POW groups, of varying experience, performed the LBBS. In order of highest skill to least skilled, the POW groups were: competitive collegiate; competitive high school; and novice. Future studies should look to specifically include only well-trained athletes when comparing the HBBS with LBBS, to minimize the dilution of results from less experienced populations.

The significant differences observed between the experienced (i.e., OLY and POW) groups and the less experienced (i.e., HBCOM and LBCOM) groups indicate that the time spent familiarizing each comparison participant with both squat styles was insufficient to create expertise in both styles before testing. The differences in joint angles between the 2 bar positions in the comparison group can also be attributed to a lack of expertise in both squat styles. Another limitation to this study was the low number of participants representing each group, as this reduced the statistical power of the model. Athletes competing at a high level were targeted to make up the experienced OLY and POW groups (i.e., international and national level, respectively). Therefore, the pool of potential participants was automatically reduced. Moreover, athletes were also recruited from different gyms in different stages of competition preparation at the time of testing. As a result of the reduced sample size, the effect size data should be carefully considered rather than interpreting the findings based on statistical significance alone. Future studies should look to compare larger cohorts of experienced HBBS and LBBS participants up to and including 100% of 1RM, with the further addition of muscle activity analysis, to complete a full profile of each squat style and improve statistical power.

Practical Applications

This study provided evidence to suggest that the LBBS is a more efficient way of squatting large loads, as demonstrated by comparable kinetic results to the HBBS, despite greater absolute loads being lifted. This study also indicates that resistance-trained individuals should not be compared/combined with well-trained athletes when comparing such a technical movement as the HBBS or LBBS, as there is an apparent influence of expertise on the performance of these techniques. With regard to training adaptations, practitioners seeking to place emphasis on the stronger hip musculature should consider the LBBS, as the greater forward lean of the movement ensures the hip muscles are engaged more so than the HBBS. It is also recommended that when the goal is to lift the greatest load possible, the LBBS may be preferable. Conversely, the HBBS is more suited to replicate movements that exhibit a more upright torso position, such as the snatch and clean, or to place more emphasis on the associated musculature of the knee joint. Future research should look to analyze the muscle activity differences between the HBBS and LBBS, up to and including 100% 1RM. The addition of this knowledge to the results presented in this study will provide a complete profile of the differences between the HBBS and LBBS.

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